$C_{Al} * M_{Al}\, \Delta T$

5 min DATA
BEFORE & AFTER
@ 30s INC.

10 cm
FOR STRING,
40W BULB

T

t

$\frac{\Delta T}{\Delta t} = 2.55 \times 10^{-3}\ \frac{°C}{s}$

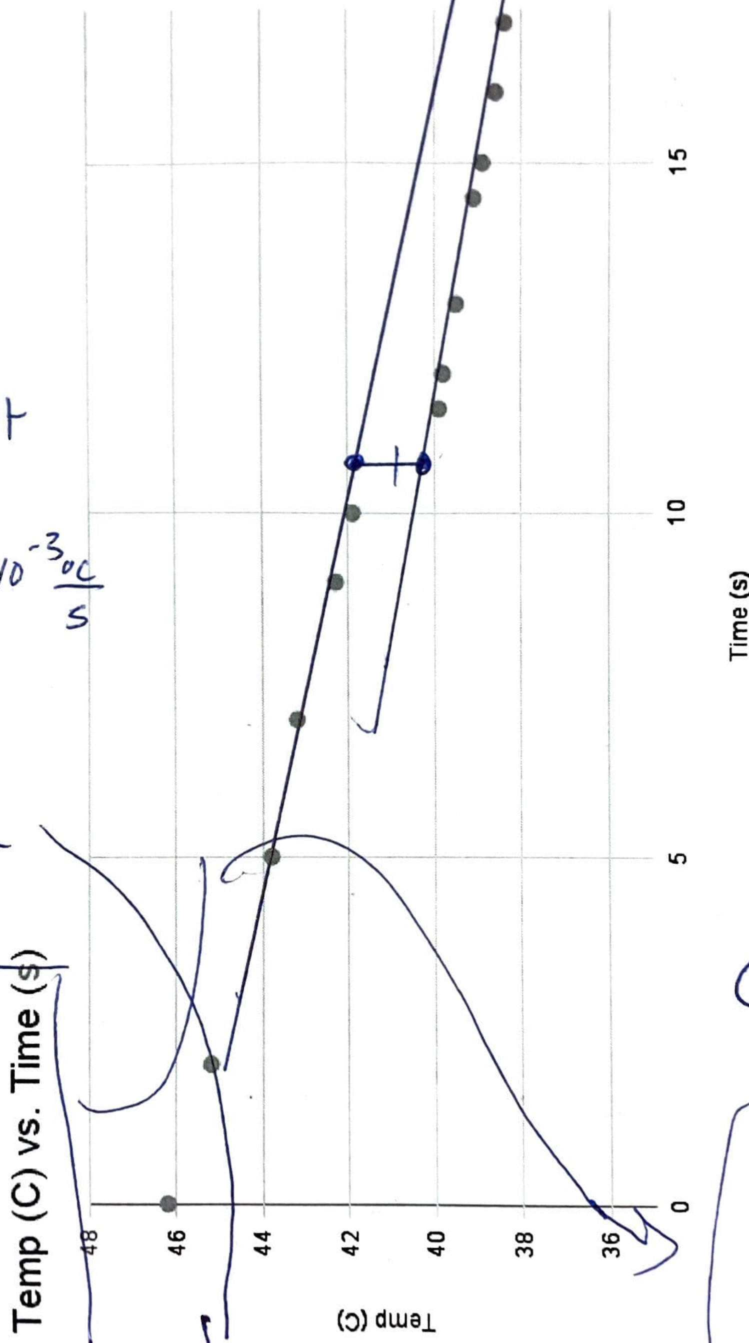
$M_{Al} = 346.5\,g$

$L \times W = 10\,cm \times 5\,cm$
$= 50\,cm^2$

FOR 5 cm

$\frac{\Delta T}{\Delta t} = 0.0111\ \frac{°C}{s}$

$50\,cm^2 \times \frac{1\,m}{100\,cm} \times \frac{1\,m}{100\,cm} =$



$0.95\,cm \times \frac{\cancel{2}8°C}{47.2\,cm}$

$\cancel{= 1.056°C}$

V_W 117.5 mL WATER

$M_W = 117.5\,g$

$M_{Al} = 18.1\,g$

$C_{Al} = 0.940$

$\frac{\Delta E}{\Delta t} = 3.460\ W$

$\rightarrow \frac{E}{\Delta t A} = 692\ \frac{W}{m^2}$

$\frac{\Delta E}{\Delta t} = \frac{C_{Al} \cdot M_{Al} \cdot \Delta T}{\Delta t} = C_{Al} \cdot M_{Al} \cdot \frac{\Delta T}{\Delta t} = 0.795\ \frac{J}{s}$

$\frac{\Delta E}{\Delta t \cdot A} = 159\ \frac{W}{m^2}$

$A_\phi = 4\pi a^2$

Time		
7:16	45.7	0
7:25	41.4	9
7:41	37.2	25
8:07	33.2	51

~~8:13~~ 8:15	46.2	0
8:17	45.2	2
8:20	43.8	5
8:22	43.2	7
8:24	42.3	9
8:25	41.9	10
	39.9	11.5
	39.8	12
	39.5	13
	39.1	14.5
	38.9	15
	38.6	16

T	~~7~~ Time (8)	t	
26.5	0		
26.6	44		
7	1.44		
8	2.20		
9	2.52		
27.0			29.0
1			29.1
2			2
3			3
4			4
5	17	38.4	5
6			6
7			7
8			8
9			9
28.0			30.0
1			30.1
2			2
3			3
4			4
5			5
6			6
7			7
8			8
9			9

time	time (s)	T (c)
0	0	26.5
0:30	30	26.5
1:00	60	26.6
1:30	90	26.7
2:00	120	26.7
2:30	150	26.9
3:00	180	26.9
3:30		26.9
4:00		27.0
4:30		27.1
500		27.2
5:30		27.3
6:00		27.3
6:30		
7:00		27.5
7:30		
8:00		27.6
8:30		
9:00		27.8
9:30		
10:00		28.0
10:30		
11:00		28.1
11:30		
12:00		28.3
12:30		
13:00		28.4
13:30		
14:00		28.6
14:30		

	Time
15:00	28.7
16	28.9
17	29.0
18	29.2
19	29.3
20	29.5
21	29.6
22	29.7
23	29.9
24	30.0
25	30.2
26	30.3
27	30.4
28	30.6
29	30.6
30	30.7
31	30.8
32	30.9
33	31.0
34	31.0
35	31.1

@ 5cm

$F_{net} = 692 \frac{W}{m^2}$

$F_{AC} = 450 \frac{W}{m^2}$

$\cancel{F_{net}} \cancel{=} F_{BULB} = 1142 \frac{W}{m^2}$

$P_{BULB} = 40W$

$\hookrightarrow F_{BULB} = \frac{40W}{4\pi(0.05\,m)^2} = 1273$

$F_{net} = 159 \frac{W}{m^2}$

$F_{AC} = \cancel{449} \cdot 450 \frac{W}{m^2}$

$F_{net} = F_{BULB} - F_{AC}$

$F_{BULB} = F_{net} + F_{AC}$

$= 600 \frac{W}{m^2}$

$P_B = 40W$

$A = 4\pi r^2$

w/ $r = 10cm$

$= 0.1257$

$\hookrightarrow F_{BULB} = 318 \frac{W}{m^2}$

$P_{BULB} = F_{BULB} \cdot A_{AC}$